

Yung Hsing (Winston) Huang

Bioinformatics Scientist/ Bioinformatician



(832) 664-1350



winstonhyh@live.com



Houston, TX



SUMMARY/OBJECTIVE

- 7+ years of academic and industrial cancer research experience
- 2+ years of large-scale spatial transcriptomics experience
- Research expertise: Tumor microenvironment, spatial transcriptomics, single-cell multiomics
- Solid background and experience in cancer biology and molecular biology techniques
- Lawful permanent resident in United States and a lawful citizen in Canada

WORK EXPERIENCE



Bioinformatician

Houston Methodist Research Institute | Houston, TX, USA | 04/2026 – Present

Spatial Omics Core, Houston Methodist Research Institute

Data analyses for Spatial Omics Core of Houston Methodist Research Institute including bulk-RNA, single-cell RNA (10X Chromium), spatial transcriptomics (10X Xenium) and spatial proteomics (Akoya Phenocycler) data, some key collaborations include:

- Integrating **Xenium** spatial transcriptomics and single-cell **T-cell repertoire sequencing** datasets from patients in our clinical collaborator's clinical trial (phase II, combined immune checkpoint inhibitor with radiation) to investigate the radiation vaccination effect on T cell clonality and immune-active niches in the tumor microenvironment of muscle-invasive bladder cancer patients (collaboration with Dr. Raj Satkunasivam)
- Leveraging **Xenium** spatial transcriptomics to characterize the niche composition and essential ligand-receptor interactions of therapy-induced senescent tumor cells that are enriched in minimal residual cancers (collaboration with Dr. Yulin Li).
- Employing in-house **Visium HD** datasets to assess cancer cell death mechanisms by comprehensively comparing the niches where tumor cells that are highly pyroptotic, senescent and/or necrotic by measures of three independent gene signatures derived by us and our collaborators (collaboration with Dr. Yulin Li and Dr. Yi-nan Gong)
- Analysis of in-house bulk RNAseq dataset from patients with kidney clear cell carcinoma to study the role of thrombosis and cancer metastasis (collaboration with Dr. Dharam Kaushik)

Keith Syson Chan Lab, Department of Urology

I participate in the following projects as an in-lab bioinformatician by providing manpower in data mining, hypothesis validation, statistical analyses and exploratory data analysis:

- Harnessing in-house **single-cell RNAseq** and **Xenium 5k** spatial transcriptomics analysis to identify heterogeneity of cancer-associated fibroblasts (CAF) in non-muscle-invasive and muscle-invasive bladder cancer patients from multiple public datasets ([Gao et al., Sci Adv, 2025](#); and a manuscript in preparation)
- Development of Caspase-1 driven pyroptosis gene signature using Cox proportional hazard (CPH) model and utilizing in-house **Xenium** spatial transcriptomics to visualize the intratumoral pyroptotic niche that results in local increase of lymphocyte-to-neutrophil ratio ([Wong et al., Nat Comm, 2026](#))
- Implementing gene signature scoring on public bulk and single-cell RNAseq datasets to validate the effect of extracellular collagen and tumor cell-expressing DDR1 on matrix-induced stress response and translation machinery (Alonzo et al. *Nature Communications*, Under Revision)
- Utilizing in-house **single-cell RNAseq** dataset prepared from patient-derived lymph node metastases to unpuzzle the fibroblast and endothelial subtypes that contribute to response of immunotherapy in bladder cancer (Results published at 2026 Keystone Symposium, Stromal Immunology by Dr. Grace L. Wong)
- Leveraging in-house **mouse single-cell RNAseq** dataset to study the effect of T cell-mediated anti-cancer immunity upon high fat diet-induced obesity in bladder cancer
- Applying in-house **mouse single-cell RNAseq** from intravenously inoculated bladder-to-lung metastasis tissue to study the relationship between B cells and NK cell activity in metastasis microenvironment.
- Employing in-house **Xenium 5k** spatial transcriptomics of non-muscle invasive bladder cancer to study the roles of tertiary lymphoid structure on disease progression.
- Mentoring two PhD students for the bioinformatics aspect of their research projects.

Bioengineer

3D Systems | Houston, TX, USA | 06/2022 – 11/2023

I worked for a subsidiary company of 3D Systems, Systemic Bio, which focused on development of

organ/tumor-on-chip development for the purposes of preclinical drug efficacy/toxicity assessment. My main job functions included:

- Assay development and optimization for samples collected from 3D organ/tumor-on-chip systems including viability/cytotoxicity assays (LDH, Cell-Titer Glo, MTS, CCK8), histology assays (cryosectioning of 3D-printed hydrogels, IHC/H&E staining), analyte quantification assays (ELISA of soluble proteins, bile acids, enzyme activity colorimetric assays) and nucleic acid assays (qPCR, PicoGreen).
Assessment of biocompatibility of organ/tumor-on-chip systems using primary human hepatocytes and immortalized human cell lines using various biomaterials, microfluidic systems and hydrogel print patterns
- Establishment of pipelines for biological/biochemical assay data collection, analyses, visualization and interpretation
Maintenance of core experimental equipment including Leica Cryostat, MiniAmp™ Plus Thermal Cycler, Nanodrop, QuanStudio 6, MACSQuant 10 Analyzer Flow Cytometer and Promega Glomax Plate Reader
- Participation of partnership million-scale funding projects using organ/tumor-on-chip for preclinical drug screening with two major pharmaceutical companies.

Postdoctoral Fellow

Houston Methodist Research Institute | Houston, TX, USA | 04/2019 – 05/2022

- Identified an innovative therapeutic target NHE6 for improved therapy efficacy in resistant multiple myeloma patients ([Huang et al., Drug Res Updates, 2026](#))
- Writing and preliminary data generation for a successfully funded CPRIT grant titled “[Targeting NHE6 to improve clinical efficacy of daratumumab in myeloma](#)” (Grant ID: RP220639)

EDUCATION / TRAINING



Master of Science (M.Sc.) in Biomedical Informatics

University of Texas Health Science Center at Houston | Houston, TX, USA | Class of 2024

- Practicum project on in silico drug screening for KMT2A::AFF1 fusion protein ([Huang et al., iScience, 2026](#))

Doctor of Philosophy (Ph.D.) in Lab Medicine and Pathology

University of Alberta | Edmonton, AB, Canada | Class of 2019

- Development of a 3D cell culture model for therapy assessment and molecular analyses of primary myeloma cells and myeloma cell lines ([Huang et al., Cancers, 2018](#); [Huang et al., Cancers, 2021](#))
- Development and preclinical assessment of anti-CD38-conjugated nanoparticles encapsulating STAT3 inhibitor as an innovative therapeutic agent for multiple myeloma ([Huang et al., Cancers, 2019](#))

Bachelor of Science, Honours Degree (BSc.(Hon.)) in Biochemistry

University of British Columbia | Vancouver, BC, Canada



PATENT

Pyroptosis-Associated Gene Signature (CASCADE) for Predicting Treatment Response in Bladder Cancer – Filed internally at Houston Methodist Research Institute # DISC2025014



RELEVANT CORE SKILLS

- Bioinformatics data processing and cleaning
- NGS analysis (bulk, scRNA, spatial omics)
- Bioinformatics databases (TCGA, GSE, etc)
- Version control and script management (GitHub)
- Pathway enrichment analyses (GO and KEGG, etc)
- Cloud computation system (SLURM)
- In vitro/vivo molecular biology techniques
- Data visualization and presentation
- Gene signature development
- Single-cell multiomics analyses (RNA, ATAC, TCR, spatial)
- Programming languages (Python and R)
- Linux for large-scale data manipulation
- In silico simulation for drug discover

RESEARCH CONTRIBUTIONS



Peer-reviewed publications

1. Chen, L.* , Cheng, F.* , **Huang, Y.-H.***, Wang, Y.* , He, J., Liu, J., Wang, Z., Lin, P., Fan, Y., Bao, L., Yi, Q., & Yang, J. (2026). NHE6-driven endosome-autophagy axis confers proteasome inhibitor resistance in multiple myeloma. Drug Resistance Updates. <https://doi.org/10.1016/j.drug.2026.101436>
2. Kumar, H.* , Tsang, S. V.* , Lee, M.-S.* , **Huang, Y.-H.***, Choi, J., Lotlikar, S. R., Song, J.-Y., Toh, A., Choi, Y.-L., Brannan, K. W., & Kim, P. (2026). FusionTarget: Computational framework for drug repurposing against modeled fusion protein structures from genomic breakpoints. iScience. <https://doi.org/10.1016/j.isci.2026.116076>

3. Wong, S. Q.* , Hayashi, K.* , Subel, E.* , Gao, H., **Huang, Y. H.**, Nikolo, F., Karabicici, M., Hoi, X. P., Alonzo, M., Kasabyan, A., Tsouko, E., Bouchier-Hayes, L., Shin, C., Porter, S., Garcia, H., Yin, Z., Wong, S. T., El-Zaatari, Z., Satkunasivam, R., Chen, C., Brooks, M., Zhou, L., Korentzelos, D., Gong, Y., Li, Y., Titus, R. & Chan, K.S. (2026). Chemotherapy-induced caspase-1 dependent pyroptosis in cancer cells remotely skews myelopoiesis to drive pro-tumorigenic systemic neutrophil-dominant inflammation. *Nat Commun* (2026). <https://doi.org/10.1038/s41467-026-71471-3>
4. Gao, H., Wong, S., Subel, E., **Huang, Y. H.**, Lee Y. C., Hayashi, K., Alonzo, M. E., Karabicici, M., Hoi, X. P., Kasabyan, A., Mo, Q., Zaatari, Z., Shen, S., Satkunasivam, R., Nikolos, F. & Chan, K. S. (2025). Caspase-1 dependent pyroptosis converts α SMA+ CAFs into collagen-IIIhigh iCAFs to fuel chemoresistant cancer stem cells. *Sci. Adv.* 11, eadt8697(2025). <https://doi.org/10.1126/sciadv.adt8697>
5. Liu, H., He, J., Bagheri-Yarmand, R., Li, Z., Liu, R., Wang, Z., Bach, D.-H., **Huang, Y. H.**, Lin, P., Guise, T. A., Gagel, R. F., & Yang, J. (2022). Osteocyte CLITA aggravates osteolytic bone lesions in myeloma. *Nature Communications*, 13(1), 3684. <https://doi.org/10.1038/s41467-022-31356-7>
6. Wang, Q., Lin, Z., Wang, Z., Ye, L., Xian, M., Xiao, L., Su, P., Bi, E., **Huang, Y.-H.**, Qian, J., Liu, L., Ma, X., Yang, M., Xiong, W., Zu, Y., Pingali, S. R., Xu, B., & Yi, Q. (2022). RAR γ activation sensitizes human myeloma cells to carfilzomib treatment through the OAS-RNase L innate immune pathway. *Blood*. <https://doi.org/10.1182/blood.2020009856>
7. Wang, Z., He, J., Bach, D.-h., **Huang, Y.-H.**, Li, Z., Liu, H., Lin, P., & Yang, J. (2022). Induction of m⁶A methylation in adipocyte exosomal lncRNAs mediates myeloma drug resistance. *Journal of Experimental & Clinical Cancer Research*. <https://doi.org/10.1186/s13046-021-02209-w>
8. **Huang, Y.-H.**, Almowaled, M., Li, J., Venner, C., Sandhu, I., Peters, A., Lavasanifar, A., & Lai, R. (2021). Three-dimensional reconstructed bone marrow matrix culture improves the viability of primary myeloma cells in-vitro via a STAT3-dependent mechanism. *Current Issues in Molecular Biology*. <https://www.mdpi.com/1467-3045/43/1/26>
9. Saqr, A., Vakili, M. R., **Huang, Y.-H.**, Lai, R., & Lavasanifar, A. (2019). Development of traceable rituximab-modified PEO-polyester micelles by postinsertion of PEG-phospholipids for targeting of B-cell lymphoma. *ACS Omega*. <https://doi.org/10.1021/acsomega.9b02910>
10. Haque, M., Li, J., **Huang, Y.-H.**, Almowaled, M., Barger, C. J., Karpf, A. R., Wang, P., Chen, W., Turner, S. D., & Lai, R. (2019). NPM-ALK is a key regulator of the oncoprotein FOXM1 in ALK-positive anaplastic large cell lymphoma. *Cancers*. <https://doi.org/10.3390/cancers11081119>
11. **Huang, Y. H.**, Vakili, M. R., Molavi, O., Morrissey, Y., Wu, C., Paiva, I., Soleimani, A. H., Sanaee, F., Lavasanifar, A., & Lai, R. (2019). Decoration of anti-CD38 on nanoparticles carrying a STAT3 inhibitor can improve the therapeutic efficacy against myeloma. *Cancers*, 11(2). <https://doi.org/10.3390/cancers11020248>
12. **Huang, Y.-H.**, Molavi, O., Alshareef, A., Haque, M., Wang, Q., Chu, M. P., Venner, C. P., Sandhu, I., Peters, A. C., Lavasanifar, A., & Lai, R. (2018). Constitutive activation of STAT3 in myeloma cells cultured in a three-dimensional, reconstructed bone marrow model. *Cancers*. <https://doi.org/10.3390/cancers10060206>
13. Wu, C., Gupta, N., **Huang, Y. H.**, Zhang, H. F., Alshareef, A., Chow, A., & Lai, R. (2018). Oxidative stress enhances tumorigenicity and stem-like features via the activation of the Wnt/ β -catenin/MYC/Sox2 axis in ALK-positive anaplastic large-cell lymphoma. *BMC Cancer*, 18(1), 1–11. <https://doi.org/10.1186/s12885-018-4300-2>
14. Soleimani, A. H., Garg, S. M., Paiva, I. M., Vakili, M. R., Alshareef, A., **Huang, Y. H.**, Molavi, O., Lai, R., & Lavasanifar, A. (2017). Micellar nano-carriers for the delivery of STAT3 dimerization inhibitors to melanoma. *Drug Delivery and Translational Research*, 7(4), 571–581. <https://doi.org/10.1007/s13346-017-0369-4>
15. Alshareef, A., Zhang, H. F., **Huang, Y. H.**, Wu, C., Zhang, J. D., Wang, P., El-Sehemy, A., Fares, M., & Lai, R. (2016). The use of cellular thermal shift assay (CETSA) to study Crizotinib resistance in ALK-expressing human cancers. *Scientific Reports*, 6(1), 1–12. <https://doi.org/10.1038/srep33710>
16. Wu, C., Zhang, H. F., Gupta, N., Alshareef, A., Wang, Q., **Huang, Y. H.**, Lewis, J. T., Douglas, D. N., Kneteman, N. M., & Lai, R. (2016). A positive feedback loop involving the Wnt/ β -catenin/MYC/Sox2 axis defines a highly tumorigenic cell subpopulation in ALK-positive anaplastic large cell lymphoma. *Journal of Hematology and Oncology*, 9(1), 1–14. <https://doi.org/10.1186/s13045-016-0349-z>